



Information Technology Institute (ITI)
Analog IC Design Track
Summer Camp 2026
Lab 03
Cascode Amplifier



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Part 1: Sizing Chart

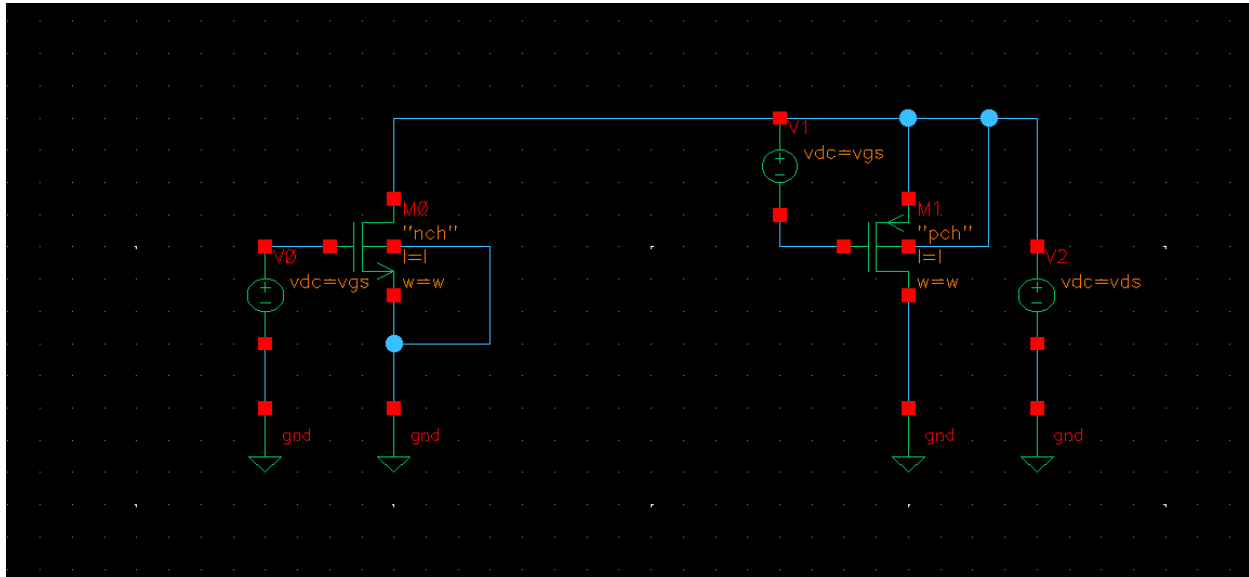


Figure 1:schematic

Using:

- $g_m * r_o = 79$ (Random)
- $I_{bias} = 49 \mu A$ (Random)
- sweep L to get intrinsic gain = 79 and $V^* = 200mV$ (sweet spot)

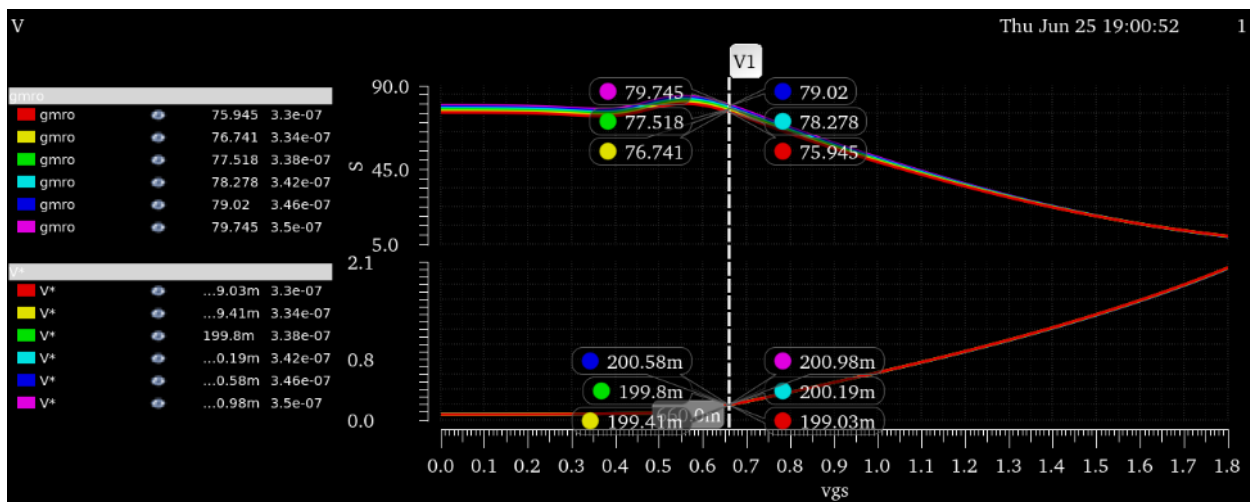


Figure 2:sweep L



From this analysis:

where $gain = 79.02$ & $V_* = 200.58\text{ mv} \rightarrow L = 0.346\text{ }\mu\text{m}$ @ $V_{gs} = 660\text{ mv}$

- Find I_{dx} @ $W=10\text{ }\mu\text{m}$

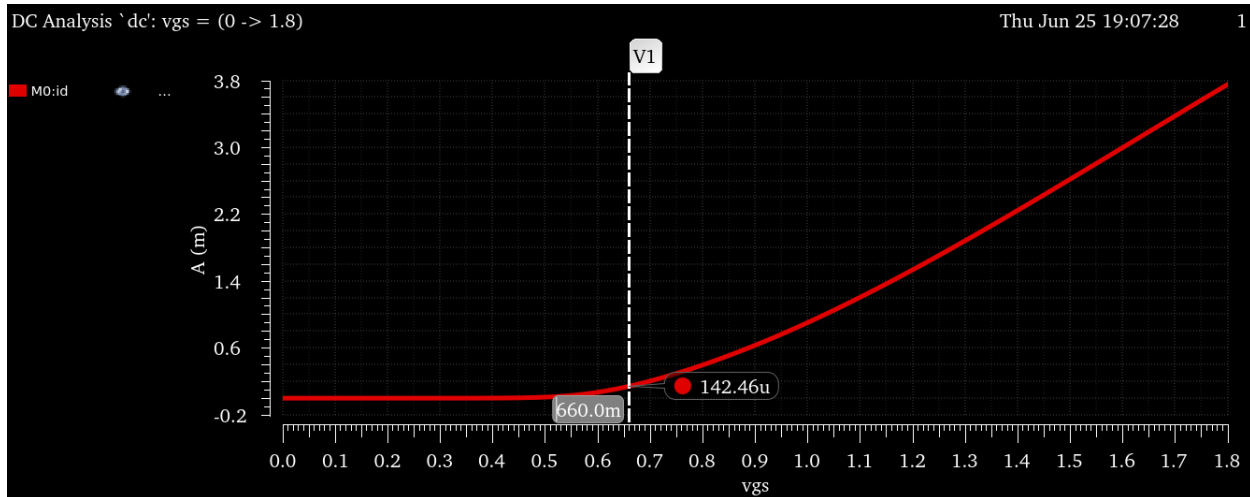


Figure 3: I_{dx} vs v_{gs}

→ Then; $I_{dx} = 142.46\text{ }\mu\text{A}$

→ Using cross multiplication $\therefore W = \frac{10 \times 49}{142.46} = 3.4395\text{ }\mu\text{m}$

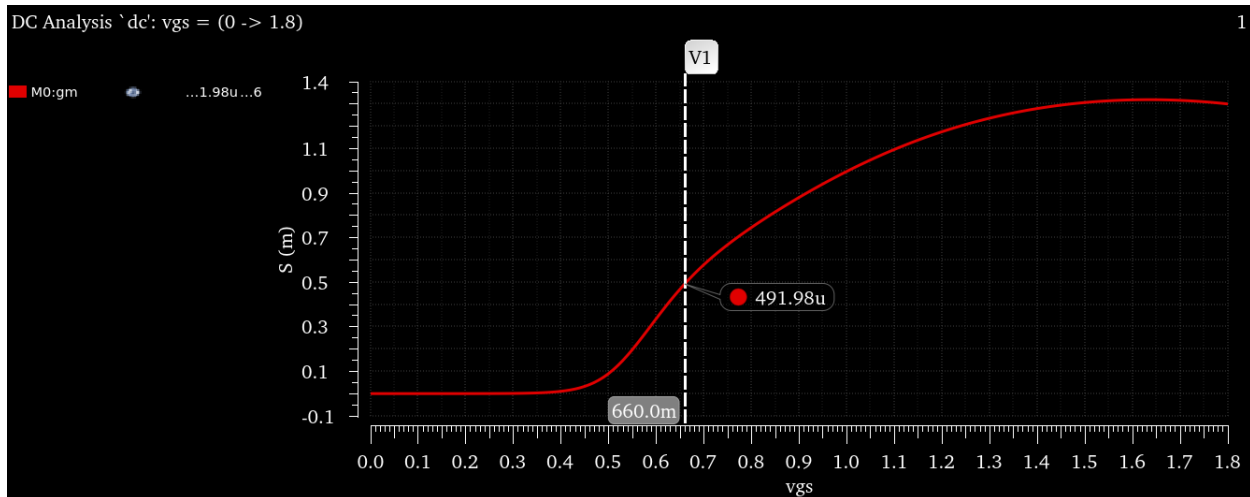


Figure 4: g_m vs V_{gs}

→ $g_m = 491.98\text{ }\mu\text{S}$ @ $L=0.346\text{ }\mu\text{m}$ & $W=3.4395\text{ }\mu\text{m}$

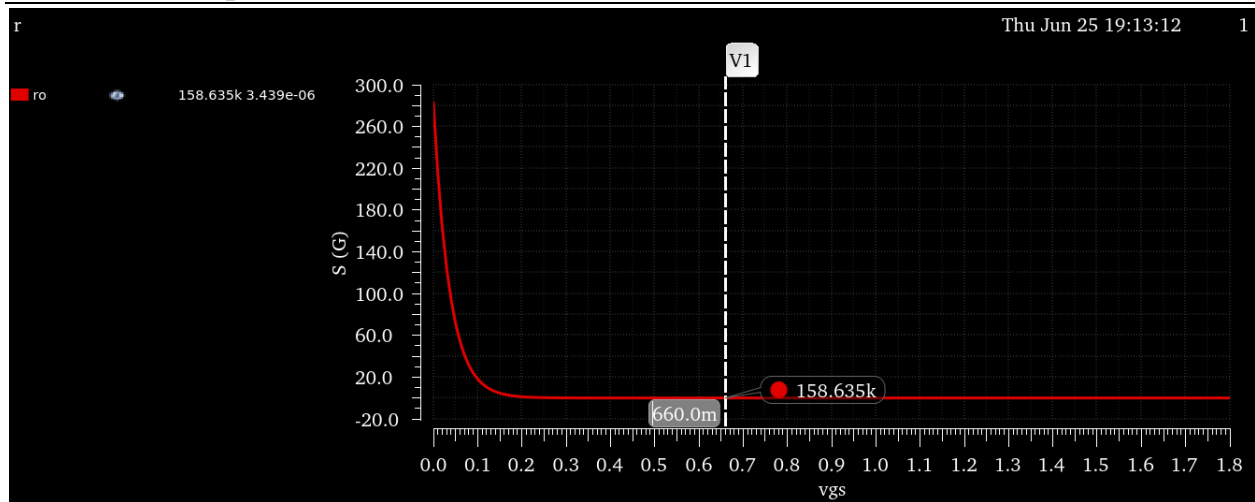


Figure 5: r_o vs v_{gs}

→ $r_o = 158.635$ k Ω

- intrinsic gain & I_d after Sizing Chart ($W = 3.4395$ μm & $L = 0.346$ μm)

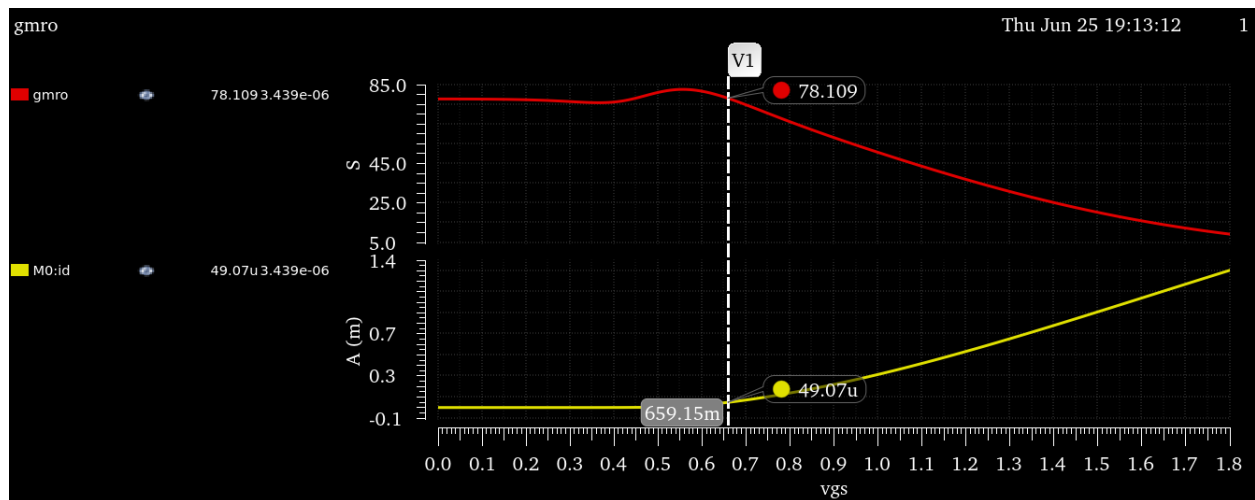


Figure 6: $g_m r_o$ & I_d after sizing

→ $g_m r_o = 78.109$

→ $I_d = 49.07$ μA



- Sizing Chart & Device Parameters

Parameter	Value
I_D	49 uA
$g_m r_o$	79
L	0.346 um
V_{GSQ}	660 mv
I_{DX}	142.46 uA
W	3.4395 um
g_m	491.98 uS
r_o	158.63 k ohm
Final Intrinsic Gain	78.02
Id	49.58 uA



PART 2: Cascode for Gain

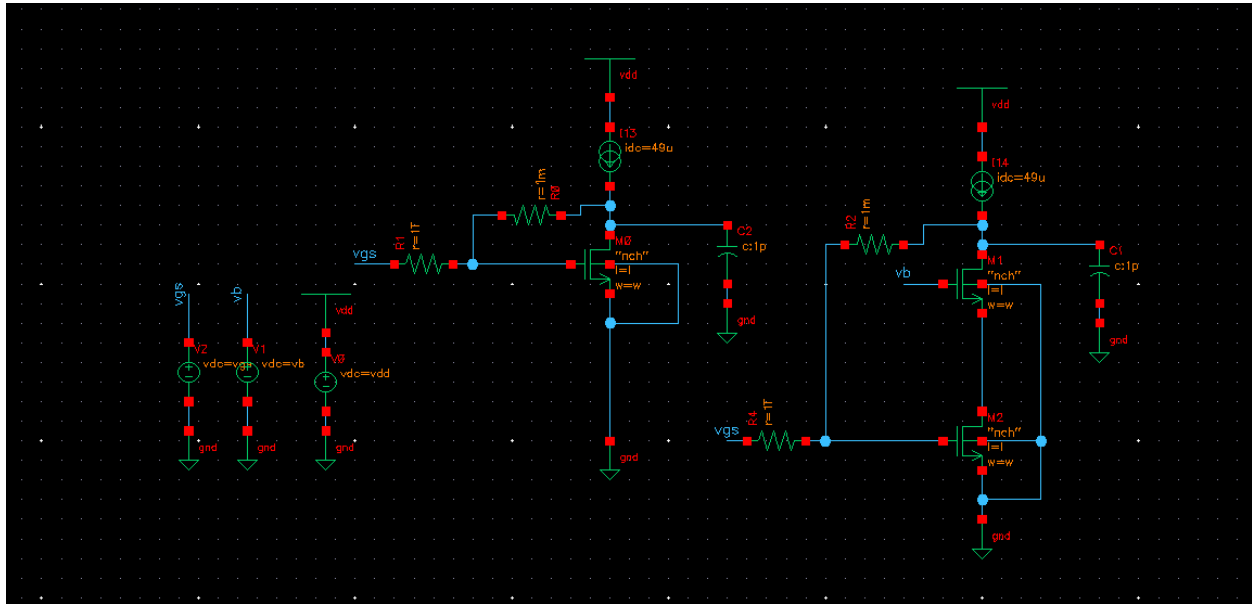


Figure 7: cascode schematic

1. OP Analysis

- Choose V_b

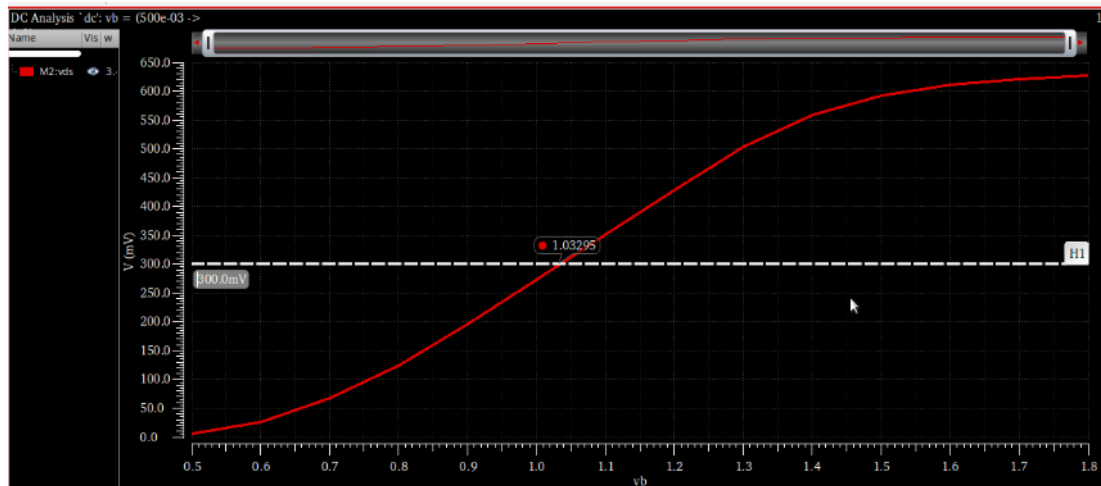


Figure 8: V_{ds} vs V_b

- Sweep V_b to reach($V_{ds} @ M2=300 \text{ mV}$) $\rightarrow V_b=1.03295 \text{ v}$



- DC OP points for M0, M1&M2

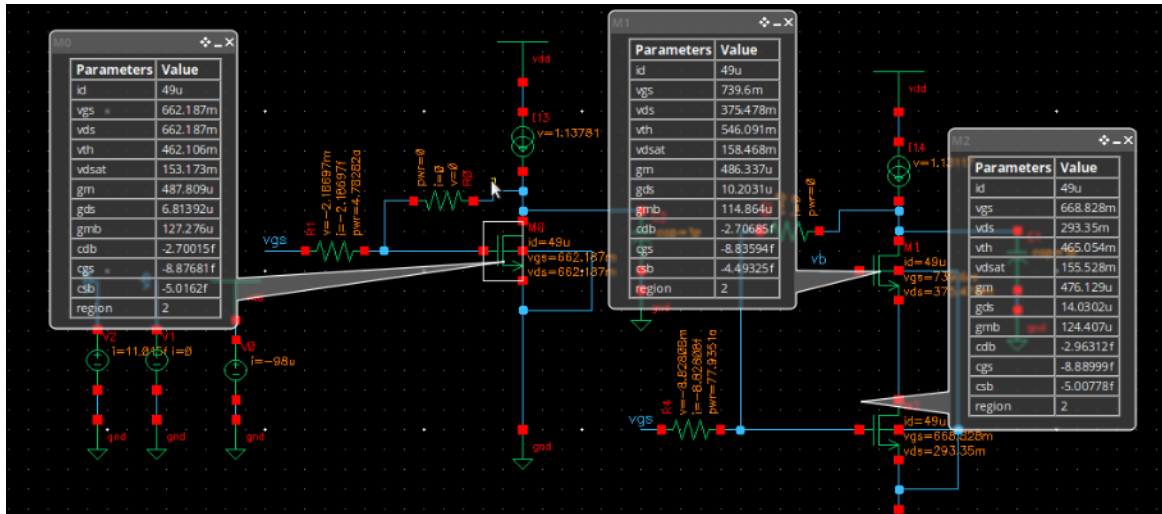


Figure 9:DC_OP

1) Operating Region Verification

Yes, all transistors are operating in the saturation region. This is verified by the simulation data where the parameter region = 2 for all devices (M0, M1, and M2). Additionally, the saturation condition $V_{ds} > V_{dssat}$ is clearly satisfied for all transistors across the design.

2) Threshold Voltage Vth Analysis

No, the transistors do not have the same threshold voltage. Transistor M1 (the cascode device) exhibits a higher threshold voltage 546 mV compared to M0 462 mV and M2 465mV. This variation is caused by the **Body Effect**; since the source of M1 is not tied to the ground, it experiences a source-to-bulk voltage $V_{sb} > 0$, which consequently increases its V_{th} .

3) Transconductance vs. Output Conductance (gm vs gds)

$$g_m \gg g_{ds}$$

In our operating point data, the transconductance g_m is in the range of 480 mS , while the output conductance g_{ds} is significantly smaller, in the range of 10 uS. This relation validates the expected high intrinsic gain (g_m/g_{ds}) of the devices in saturation.



4) Gate vs. Body Transconductance (g_m vs g_{mb})

$$g_m > g_{mb}$$

Based on the extracted parameters, g_m is 480 μS , while the body transconductance g_{mb} is 120 μS . As expected in standard MOSFET operation, the transconductance due to the body effect is only a fraction (typically 10% to 30%) of the main gate transconductance.

5) Gate Capacitances (C_{gs} vs C_{gd})

$$|C_{gs}| \gg |C_{gd}|$$

Because the transistors are biased in the saturation region, the channel is pinched off at the drain side. This makes the gate-to-drain capacitance C_{gd} very small (primarily just the overlap capacitance), whereas the gate-to-source capacitance C_{gs} models the majority of the active channel charge, making it significantly larger ($\approx \frac{2}{3} C_{ox} WL$).

6) Junction Capacitances (C_{sb} vs C_{db})

$$|C_{sb}| > |C_{db}|$$

Ignoring the negative sign convention used by the SPICE simulator (which merely indicates current/charge direction), the absolute magnitude of the source-to-bulk capacitance C_{sb} is approximately 5 fF, which is demonstrably greater than the drain-to-bulk capacitance C_{db} sitting at approximately 2.7 fF.



2. AC Analysis

- DC gain, BW, GBW

Test	Output	Nominal
cascode_amp:cascode:1	dB20(VF("/vout1"))	
cascode_amp:cascode:1	ymax(dB20(VF("/vout1")))	37.1
cascode_amp:cascode:1	ymax(mag(VF("/vout1")))	71.59
cascode_amp:cascode:1	bandwidth(VF("/vout1") 3 "low")	1.081M
cascode_amp:cascode:1	gainBwProd(VF("/vout1"))	77.55M
cascode_amp:cascode:1	dB20(VF("/vout2"))	
cascode_amp:cascode:1	ymax(dB20(VF("/vout2")))	66.17
cascode_amp:cascode:1	ymax(mag(VF("/vout2")))	2.034k
cascode_amp:cascode:1	bandwidth(VF("/vout2") 3 "low")	36.28k
cascode_amp:cascode:1	gainBwProd(VF("/vout2"))	73.96M

- Bode plot of CS and cascode
- Vout1 → CS (Red)
- Vout2 → cascode (Yellow)

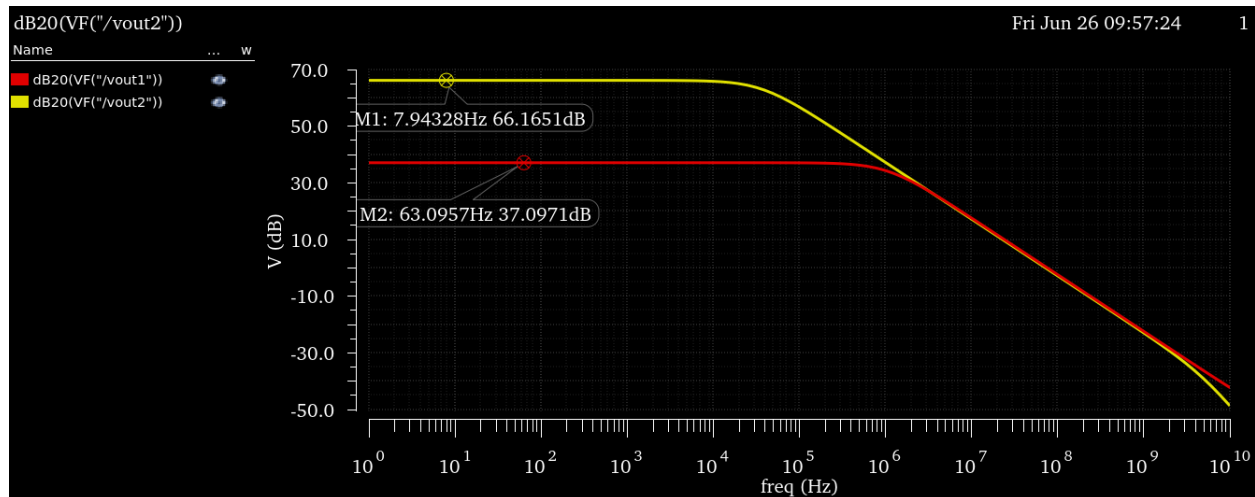


Figure 10: Bode plot cs vs cascode

- Hand analysis

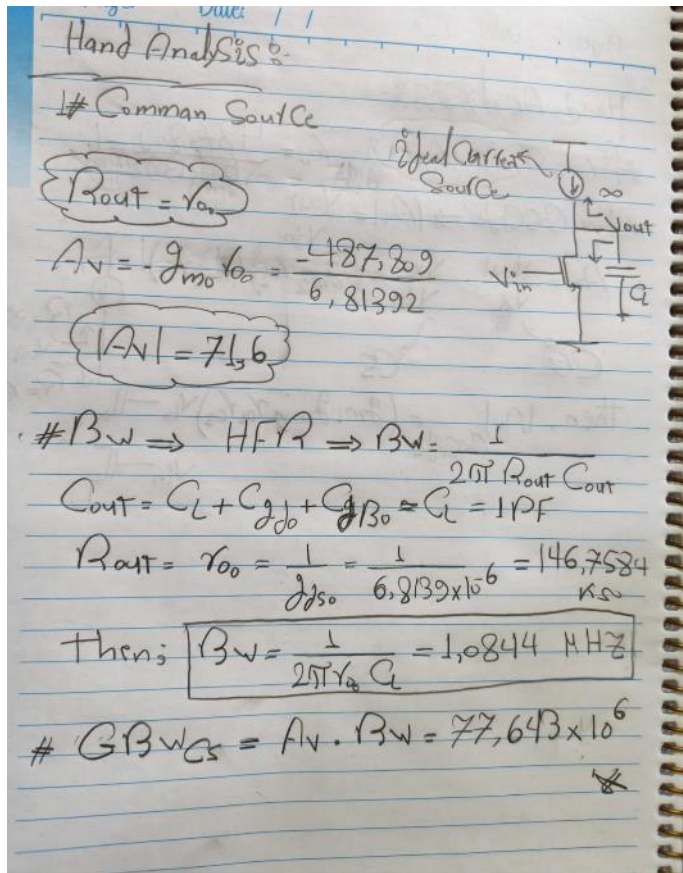


Figure 12: CS hand analysis

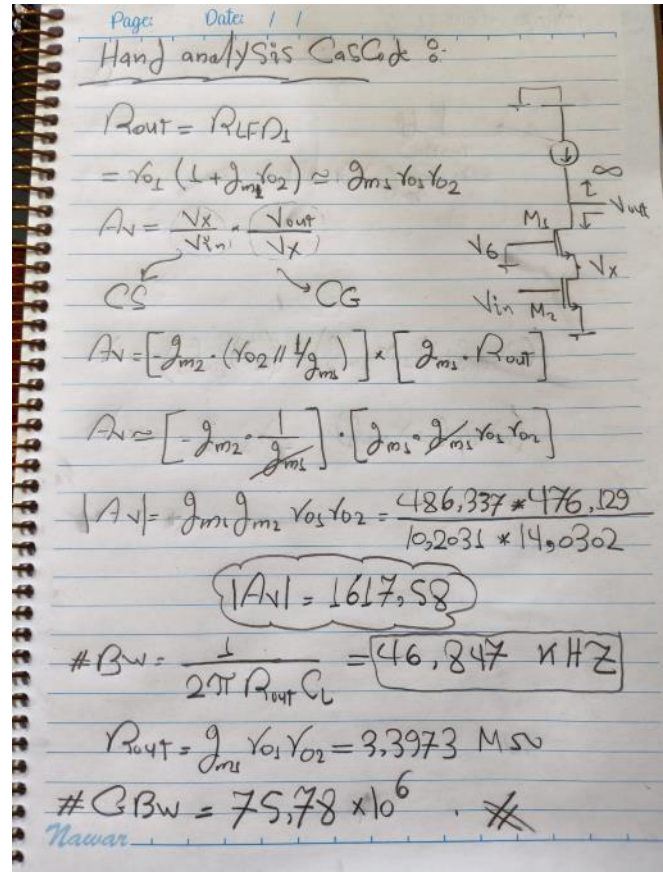


Figure 11: cascode Hand analysis



- **Comparing simulation and hand analysis.**

Parameter	CS Amp (Simulated)	CS Amp (Calculated)	Cascode Amp (Simulated)	Cascode Amp (Calculated)
DC Gain	71.59 V/V	71.6 V/V	2034 V/V	1617.58 V/V
Bandwidth	1.081 MHz	1.0844 MHz	36.28 kHz	46.847 kHz
GBW	77.55 MHz	77.643 MHz	73.96 MHz	75.78 MHz

- **Comment on the results:**

As observed in the comparison table, the hand analysis for the Common-Source (CS) amplifier closely matches the simulation results. However, for the Cascode amplifier, there is a noticeable deviation in the DC gain (and inversely, the bandwidth) between the calculated and simulated values. This discrepancy is expected; manual calculations rely on simplified models that do not fully account for short-channel effects, body effect variations, and the non-ideal scaling of output resistance r_O present in the complex BSIM models used by the simulator. Despite this deviation in gain, the Gain-Bandwidth Product (GBW) remains remarkably consistent (75 MHz) across both methods and both topologies. This perfectly validates the theoretical trade-off: the cascode configuration significantly boosts the DC gain by increasing the overall output resistance, which in turn reduces the dominant-pole bandwidth proportionally, leaving the GBW dictated primarily by the input transconductance and load capacitance constant.